ASTRONOMISCHES RECHEN-INSTITUT
HEIDELBERG

Simulations of dissolving star clusters in the Galactic Center

Andreas Ernst[1,2]

[1]Astronomisches Rechen-Institut, Zentrum für Astronomie der Universität Heidelberg, Germany
[2]Max-Planck-Institut für Astronomie Heidelberg

Computing and Simulation in Astronomy and Physics
UMassD Physics Department, November 14-15, 2009

The group

- Prof. Dr. Rainer Spurzem (presently NAOC/KIAA, Beijing)
- Privatdozent Dr. Andreas Just
- Dr. P. Berczik, Dr. J. Fiestas
- PhD / Master students

Collaboration

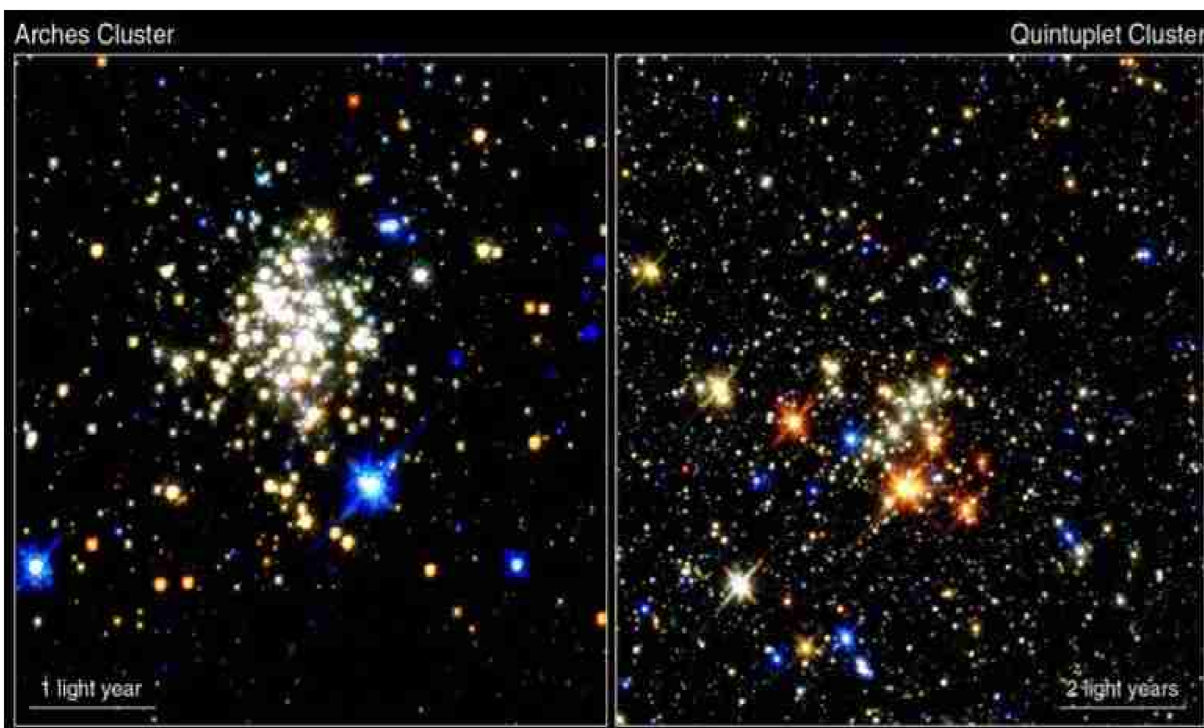
- Dr. Sverre Aarseth, Cambridge, UK

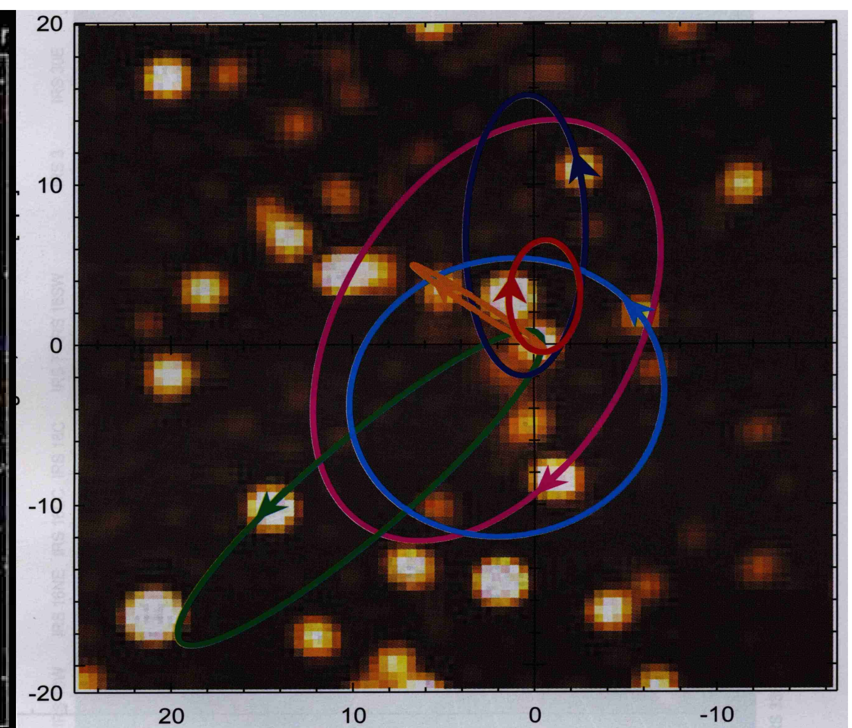
Outline

- Motivation: The “Paradox of youth“
- The Cluster Inspiral Scenario
- New *N*-body program
- The European Grid (DEISA)
- Results / Theory
- Conclusions

The Paradox of Youth (I)

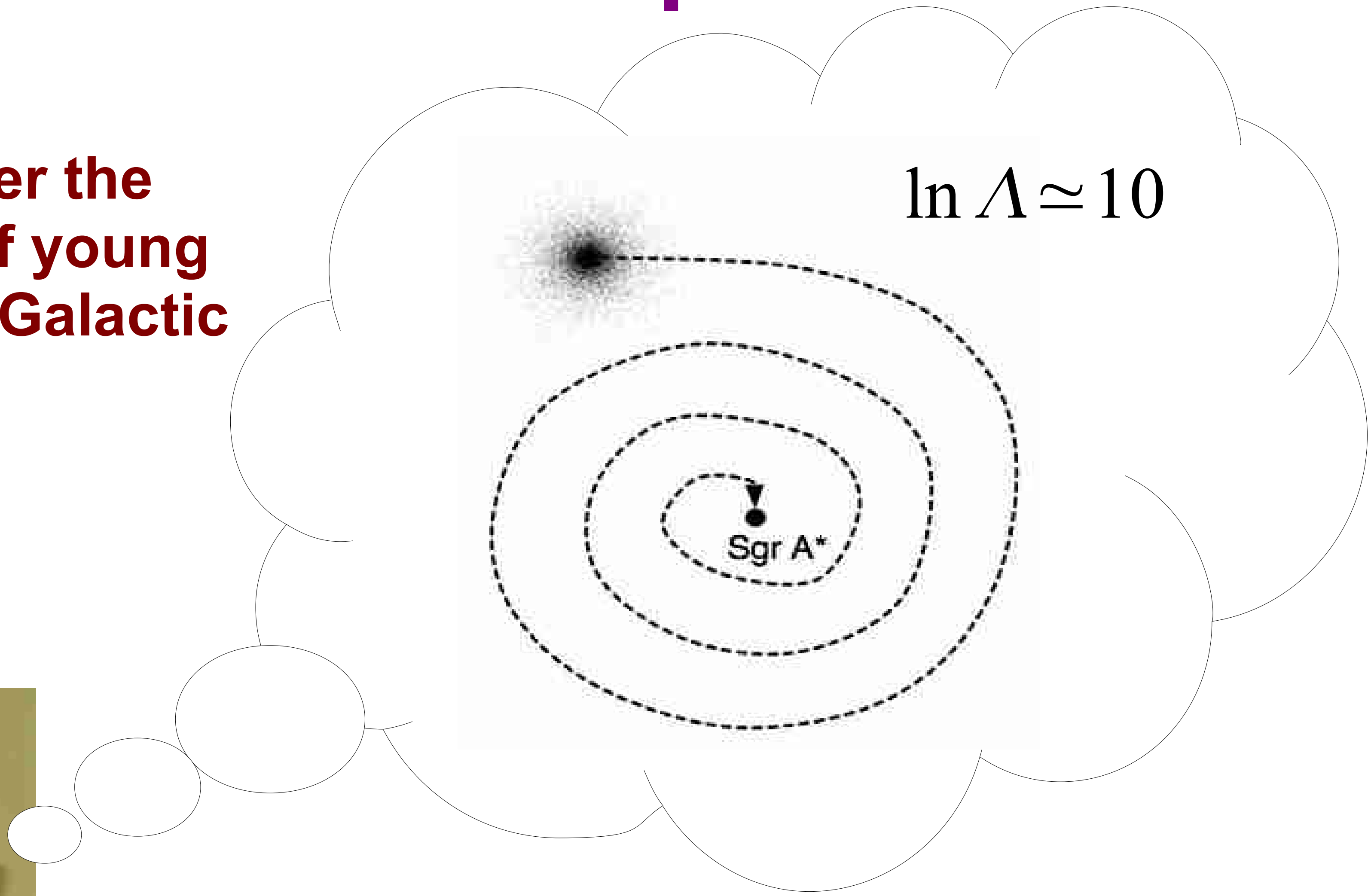
- **Young stars** (a few Myr old!) have been observed in the Galactic Center
  - In **compact comoving groups** (R_g<1 pc) IRS 13, IRS 16 SW (Maillard et al. 2004, Lu et al. 2005)
  - Many **individual massive OB stars** (R_g < 1 pc) (Paumard et al. 2006)
  - In young star clusters: **Arches** and **Quintuplet**





The star cluster inspiral scenario

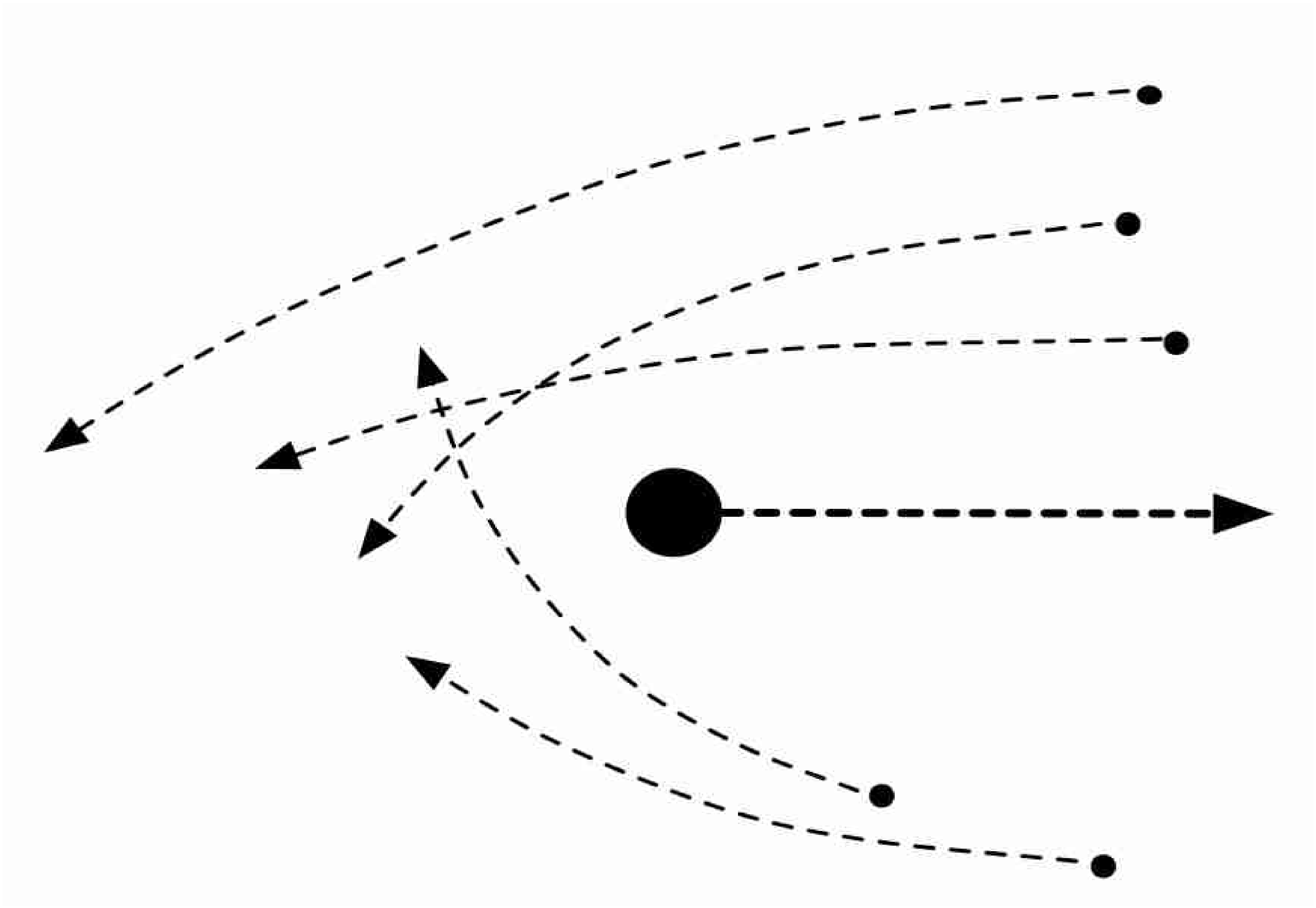
10 years after the discovery of young stars in the Galactic Center...



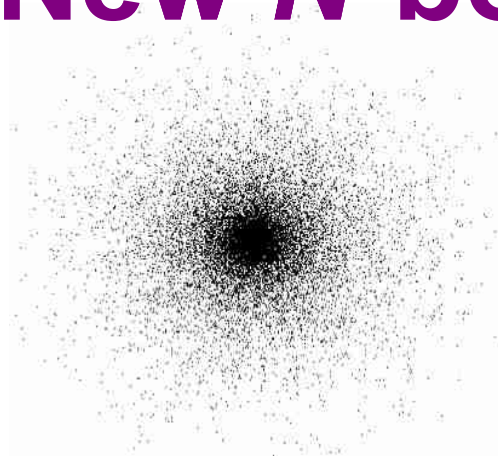
O. Gerhard, Ap. J. 546, L39-L42 (2001)

Similar ideas: Capuzzo-Dolcetta (1990s)

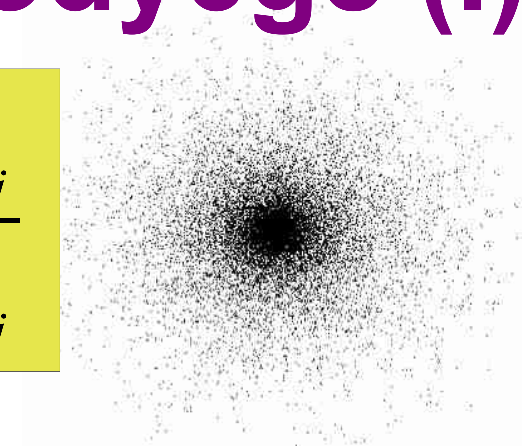
Dynamical friction



New *N*-body program: nbody6gc (I)



$$\ddot{\vec{r}}_i = -G \sum_{j=1\,(\neq i)}^{N} m_j \frac{\vec{r}_{ij}}{r_{ij}^3}$$

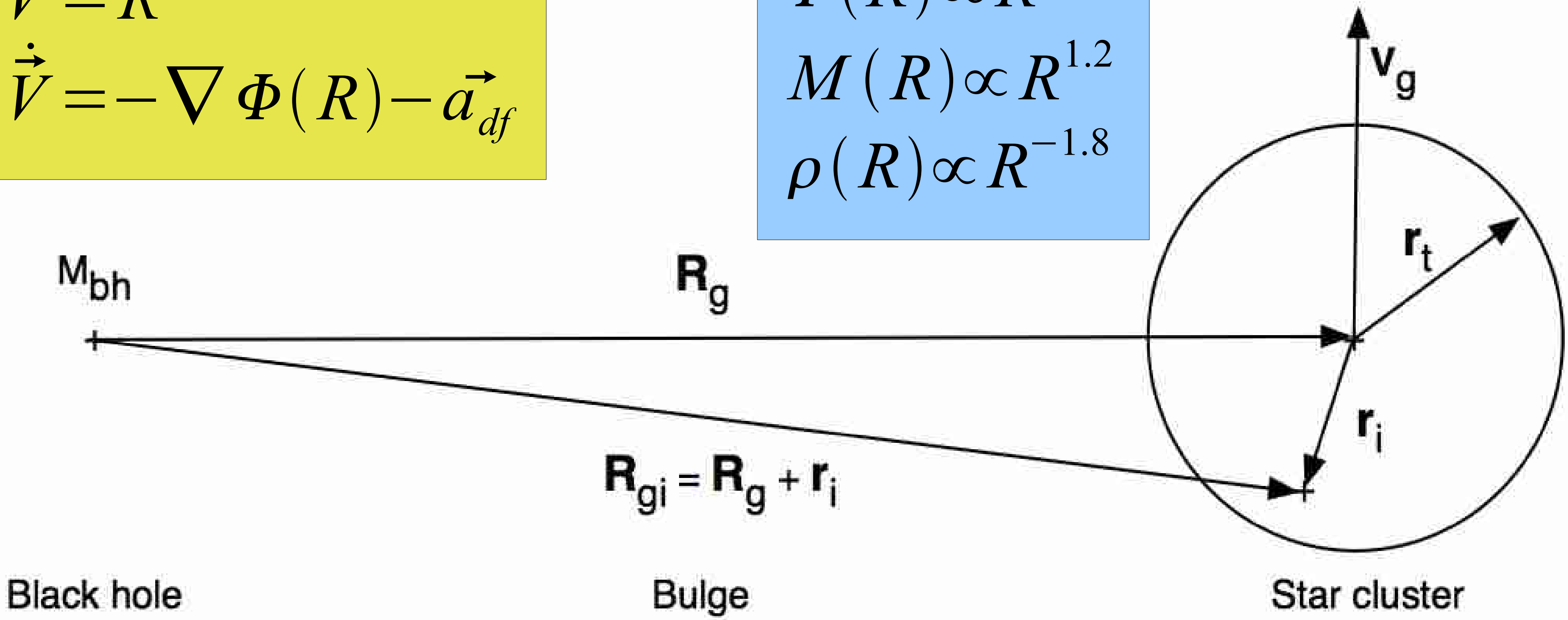


- Highly efficient codes: **nbody6** (Aarseth 1999, 2003), **nbody6++** (parallel, Spurzem 1999)
- 3*N* Newtonian equations of motion are solved using a 4th-order Hermite scheme
- **KS Regularization** (Kustaanheimo & Stiefel 1965) / **Chain regularization** (Mikkola & Aarseth 1990,1993,1996,1998)
- Different physical processes occur: Core collapse, Mass segregation, Escape of stars, Stellar evolution

New *N*-body program: nbody6gc (II)

$$\vec{V} = \dot{\vec{R}}$$
$$\dot{\vec{V}} = -\nabla \Phi(R) - \vec{a}_{df}$$

$$\Phi(R) \propto R^{0.2}$$
$$M(R) \propto R^{1.2}$$
$$\rho(R) \propto R^{-1.8}$$



- Equations of motion of the star cluster in the potential of a **black hole** and the **Galactic bulge** is solved using an **8th-order composition scheme**
- ***A full 3D tidal force*** from the **black hole** and the **Galactic bulge** acts on the cluster stars (**no approximations!**)

New *N*-body program: nbody6gc (III)

Dynamical Friction:

$$\frac{d\vec{V}}{dt} = -\frac{4\pi \ln \Lambda G^2 \rho M}{V^2}\left[\operatorname{erf}\left(\frac{V}{V_c}\right) - \frac{2}{\sqrt{\pi}}\exp\left(-\frac{V^2}{V_c^2}\right)\right]\frac{\vec{V}}{V}$$

Chandrasekhar, Ap. J. 97, 255 (1943)

Coulomb logarithm:

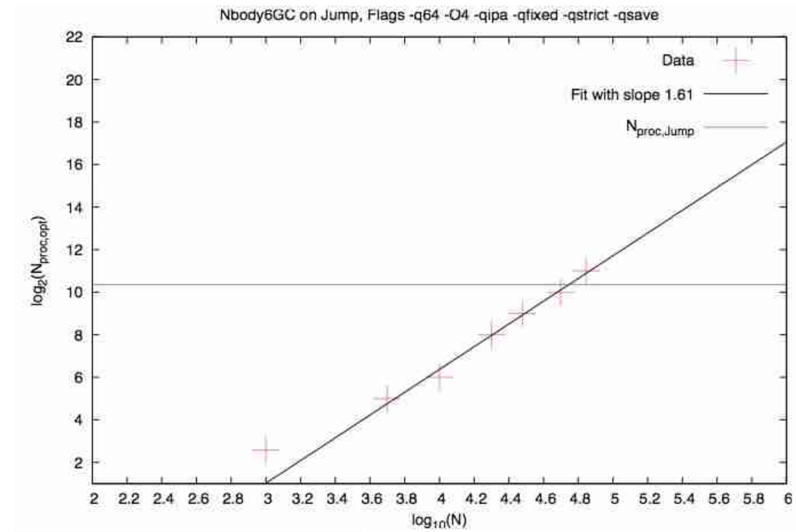
$$\ln \Lambda = \ln\left(\frac{b_1}{b_0}\right), \qquad b_0 = r_h, \qquad b_1 \simeq \frac{\nabla \rho}{\rho}$$

Just & Penarrubia (2005)

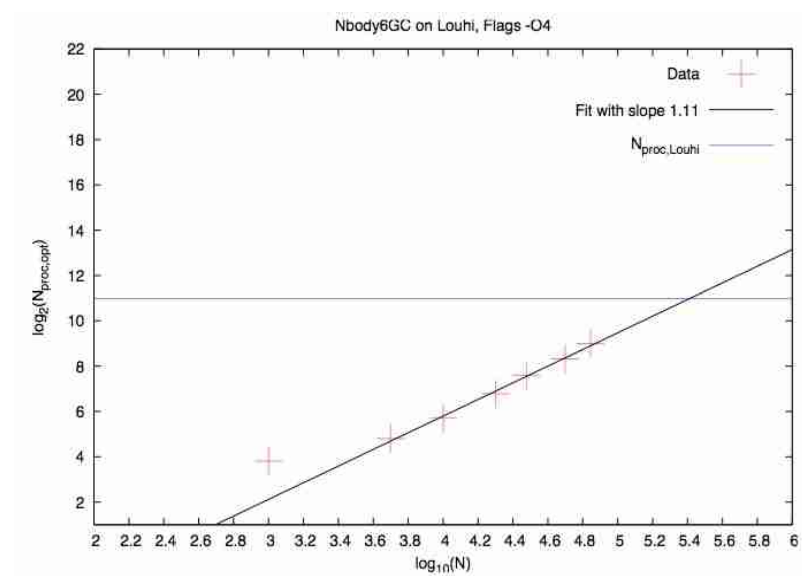
DEISA Grid Initiative

“**D**istributed **E**uropean **I**nfrastructure for **S**upercomputing **A**pplications“

Jump, Research Center Jülich (Power 4+)

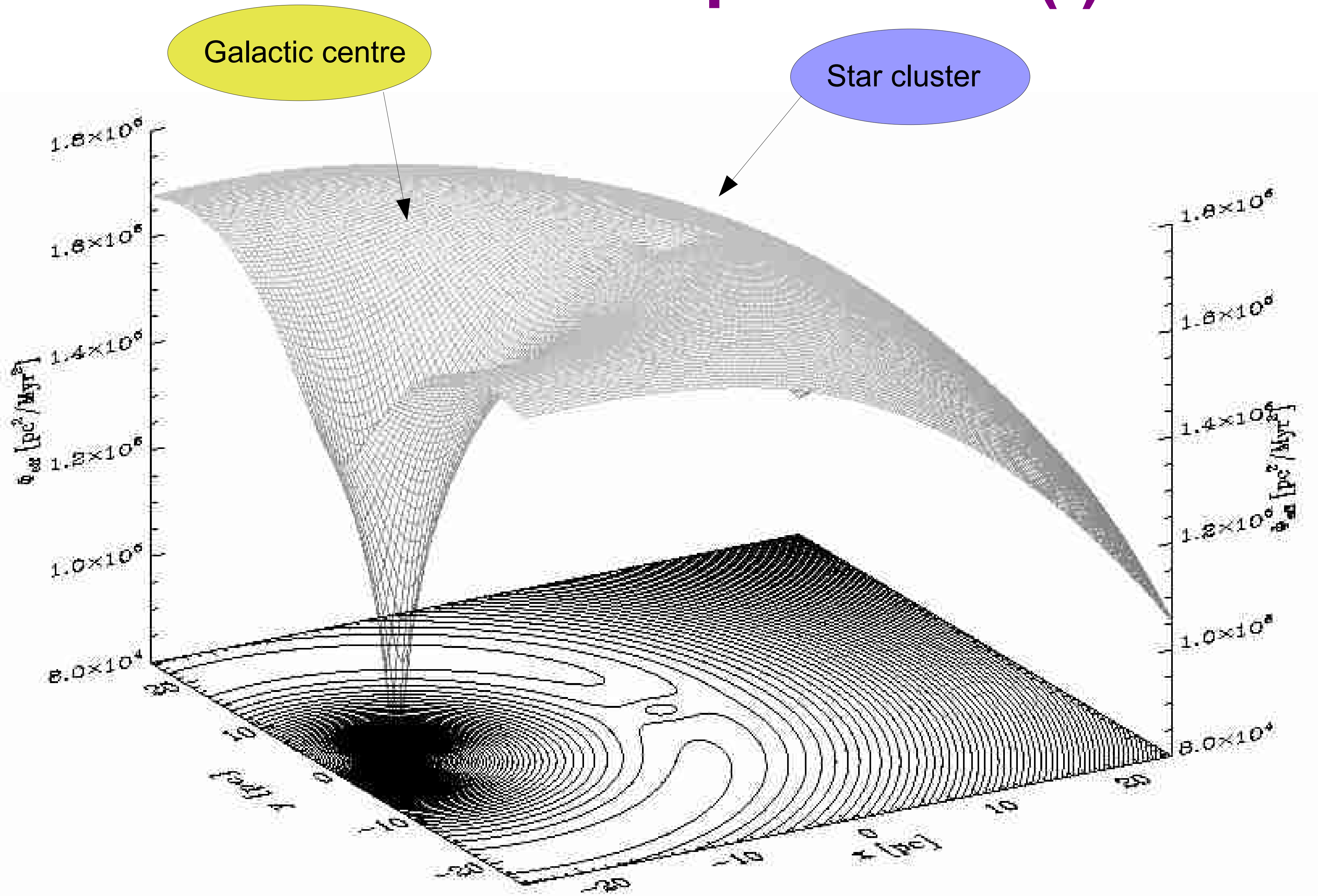


Louhi,
CSC Helsinki
(Cray XT4)

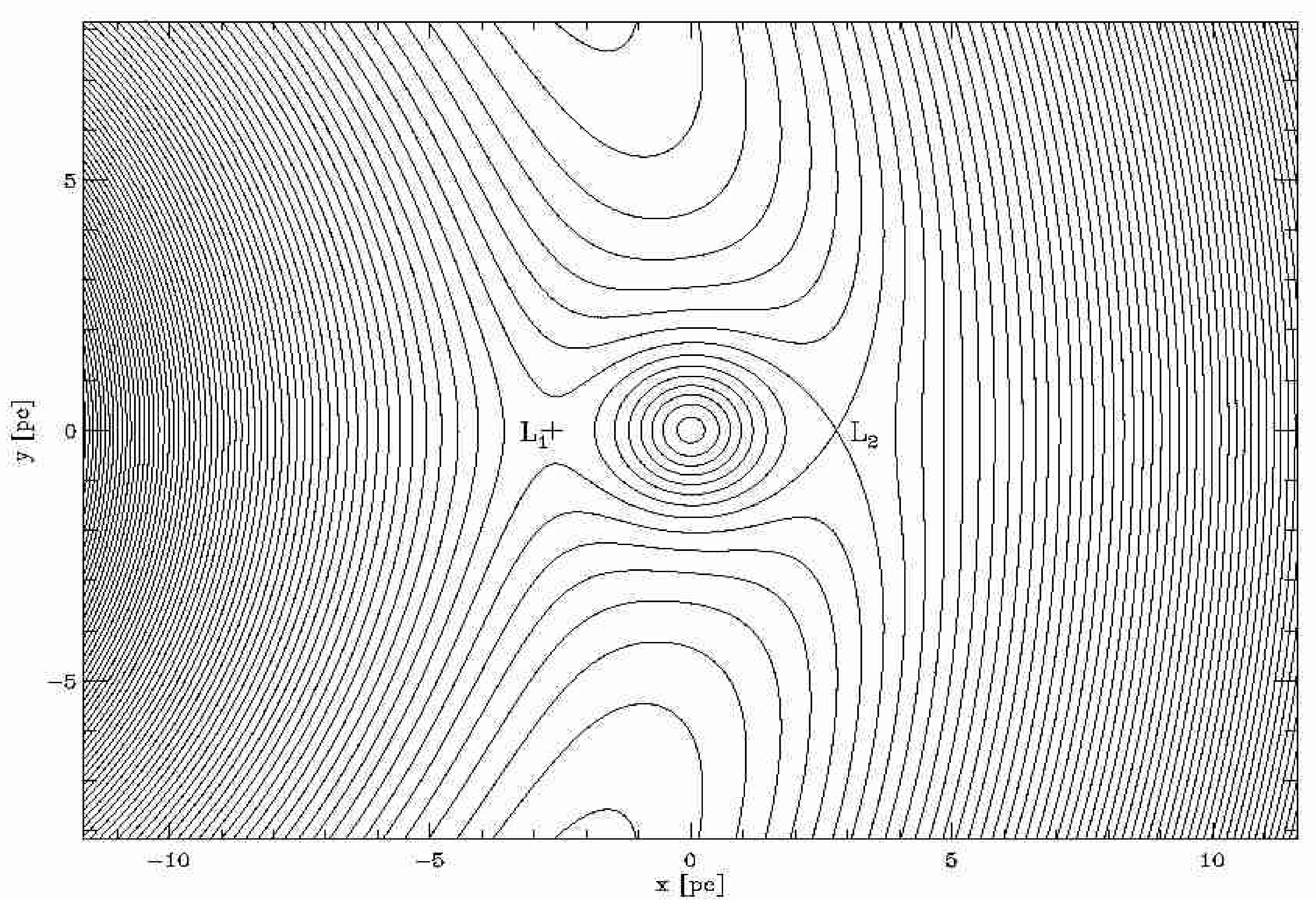


Other sites we use: HPCx Edinburgh (Power 5),
SARA Amsterdam (Power 5),
CINECA Bologna (Power 5)

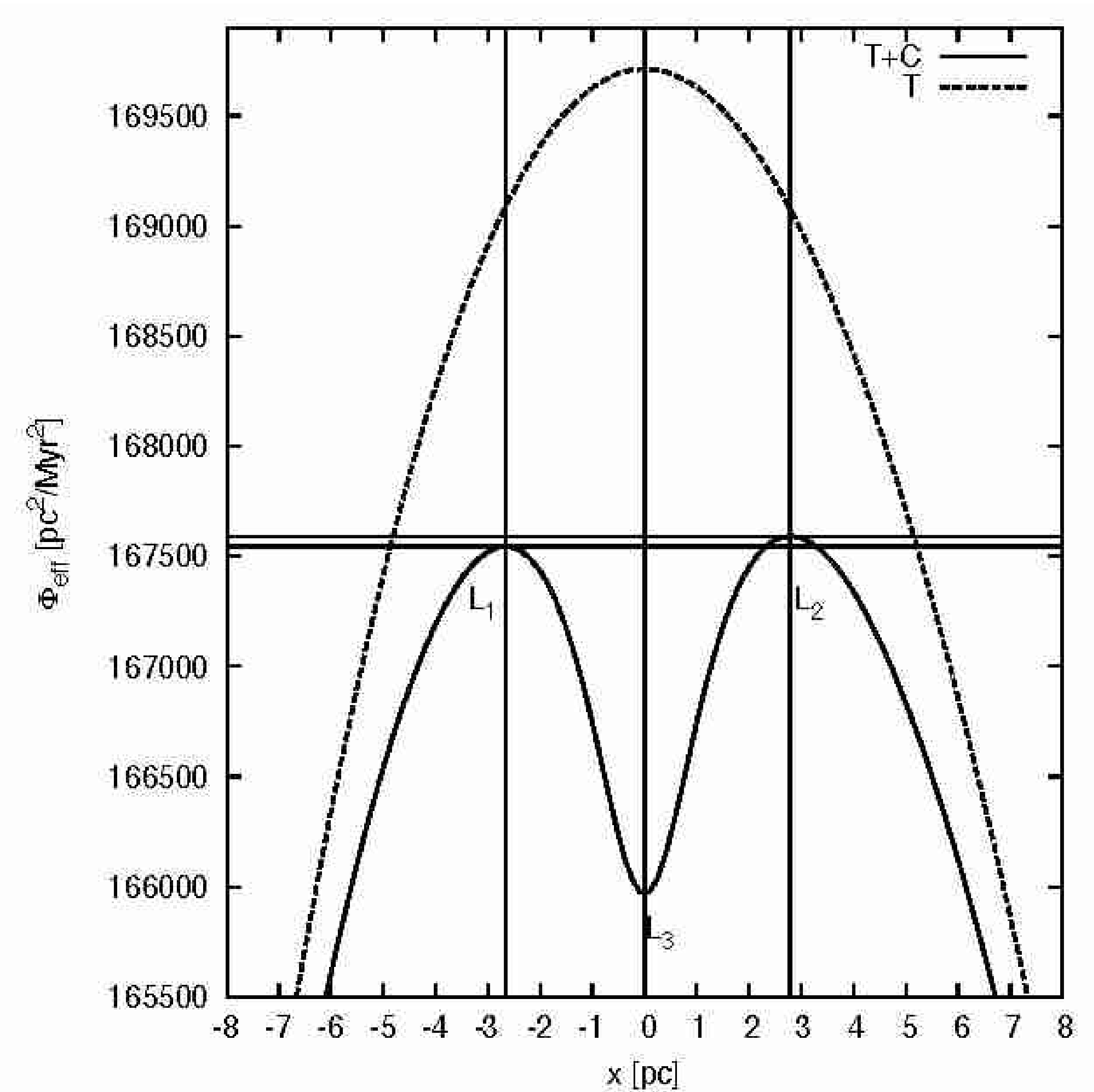
The effective potential (I)



The effective potential (II)

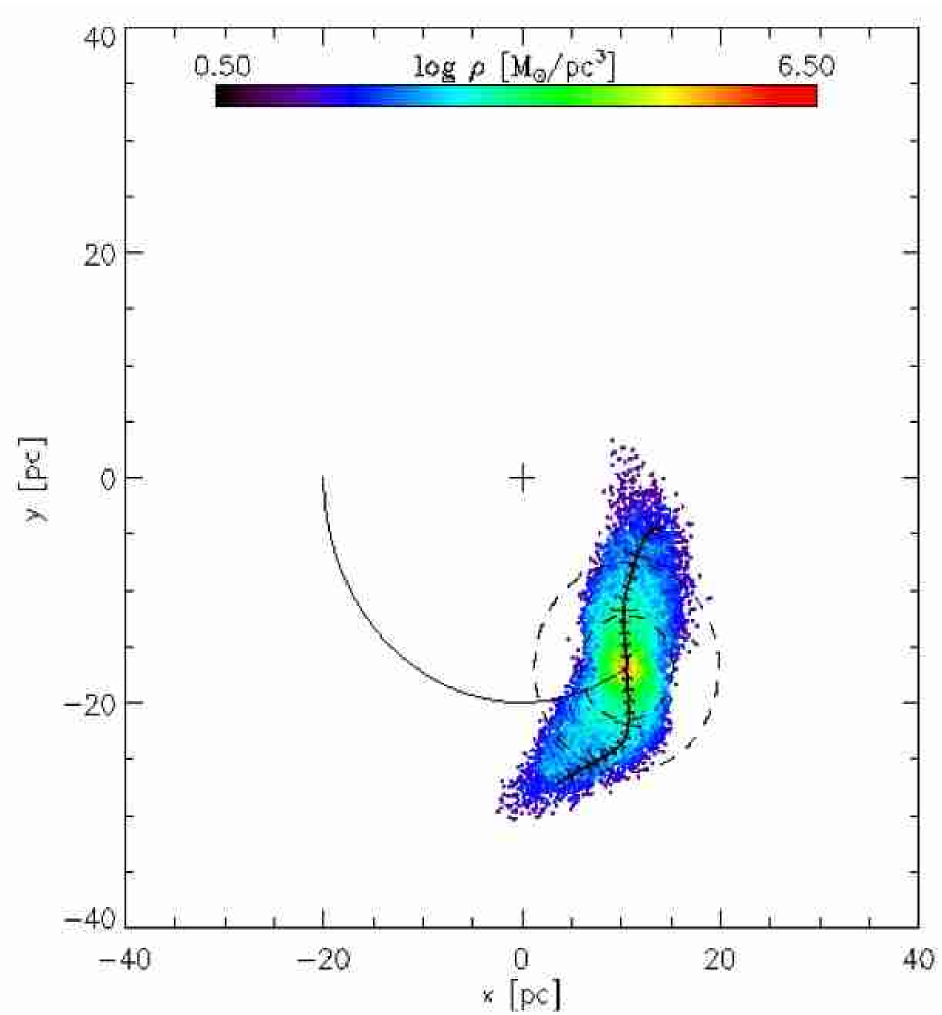


The effective potential (III)

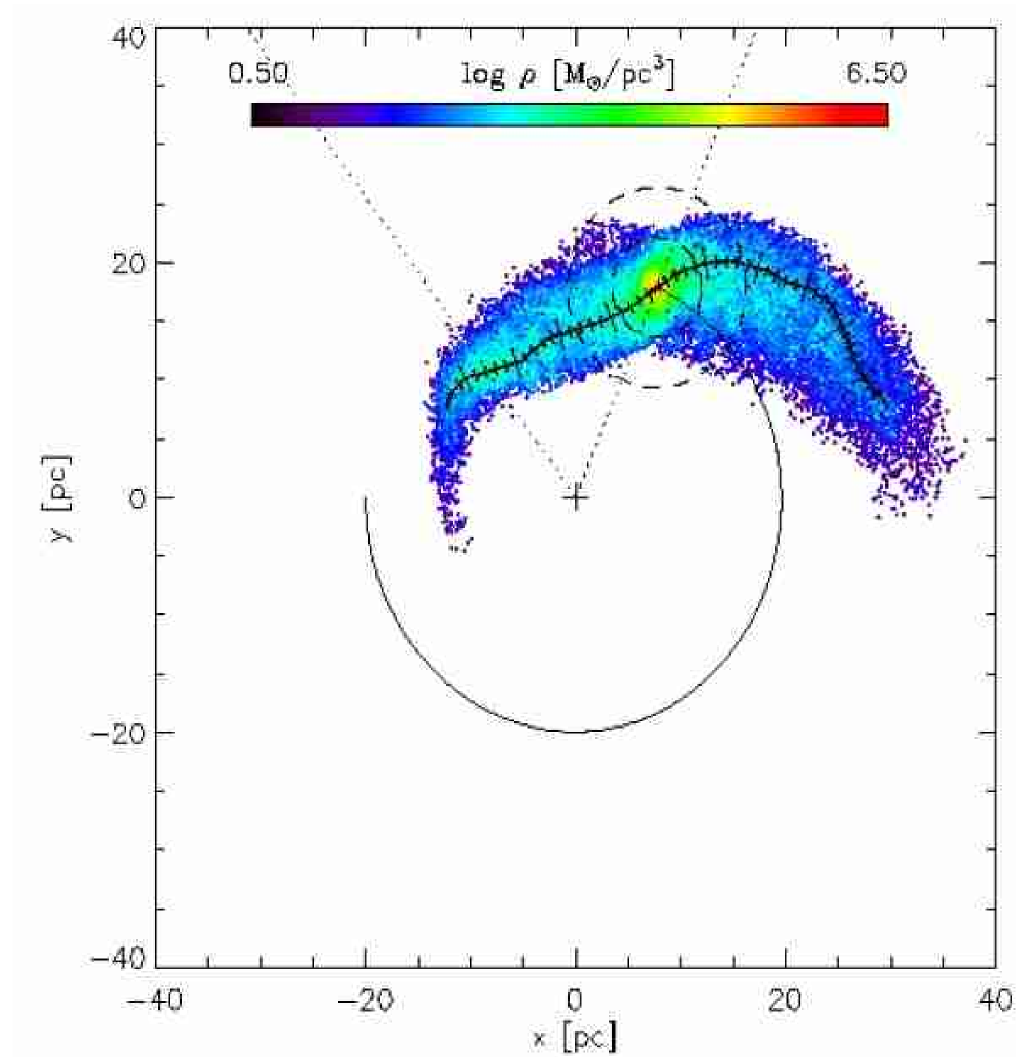


- Parabola-like effective tidal potential
- L_1/L_2 no longer symmetric w.r.t. the cluster center
- L_1/L_2 at different energies

Dissolution (I)



t=0.22 Myr

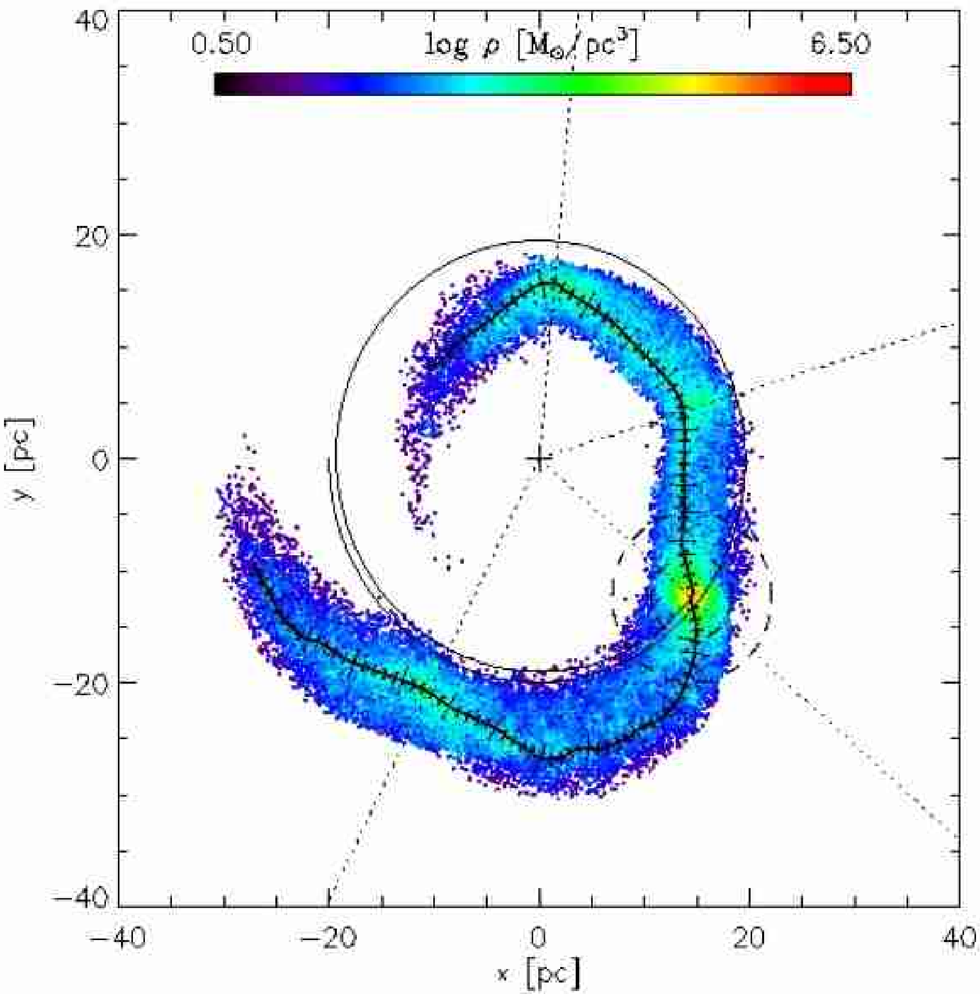


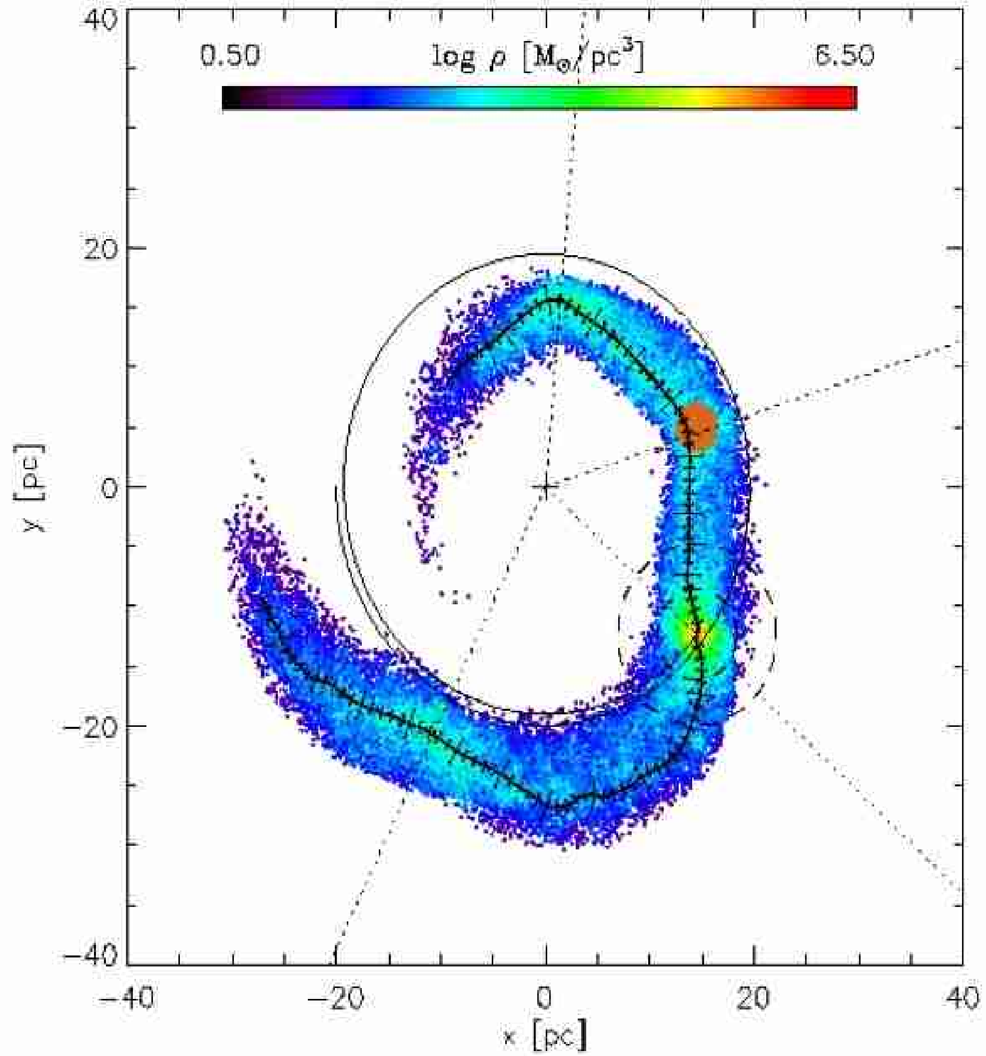
t=0.43 Myr

Tidal arms form!

Dissolution (II)

t=0.87 Myr



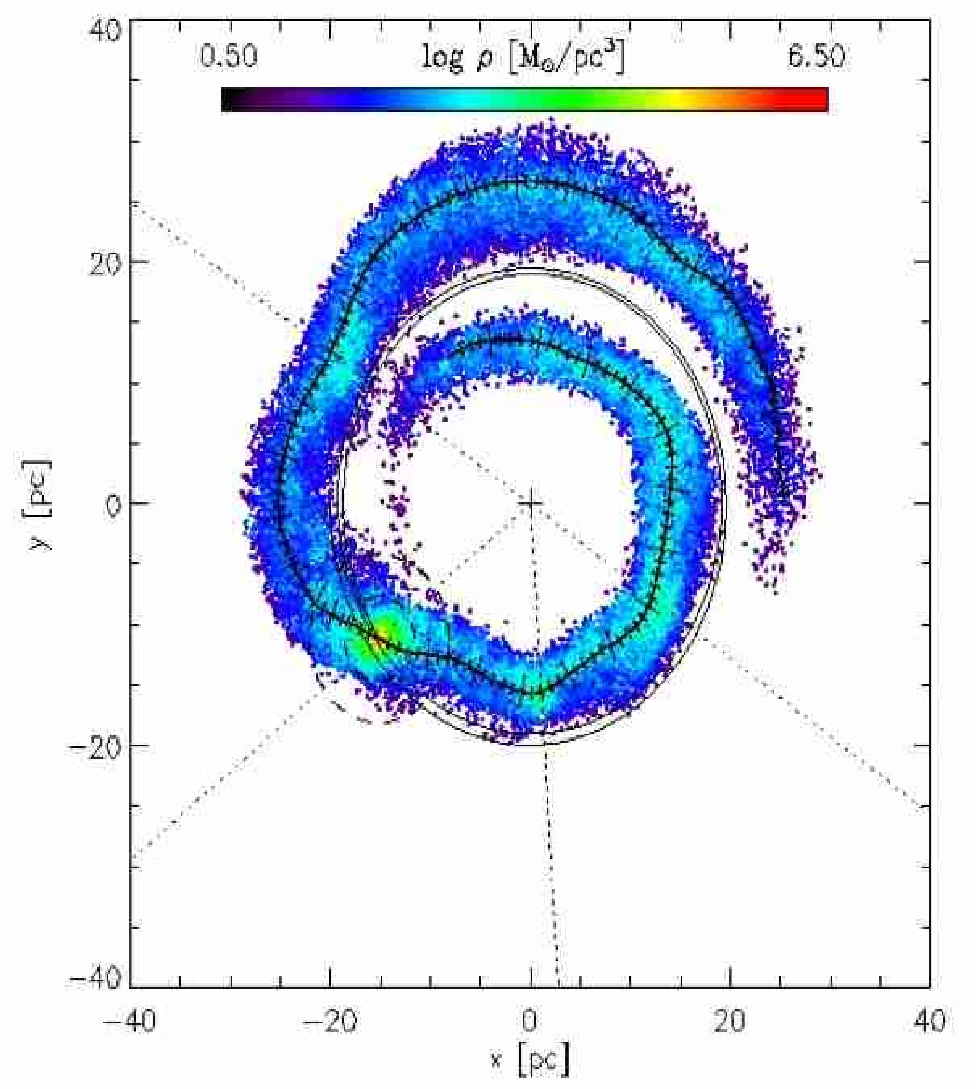


Clumps!

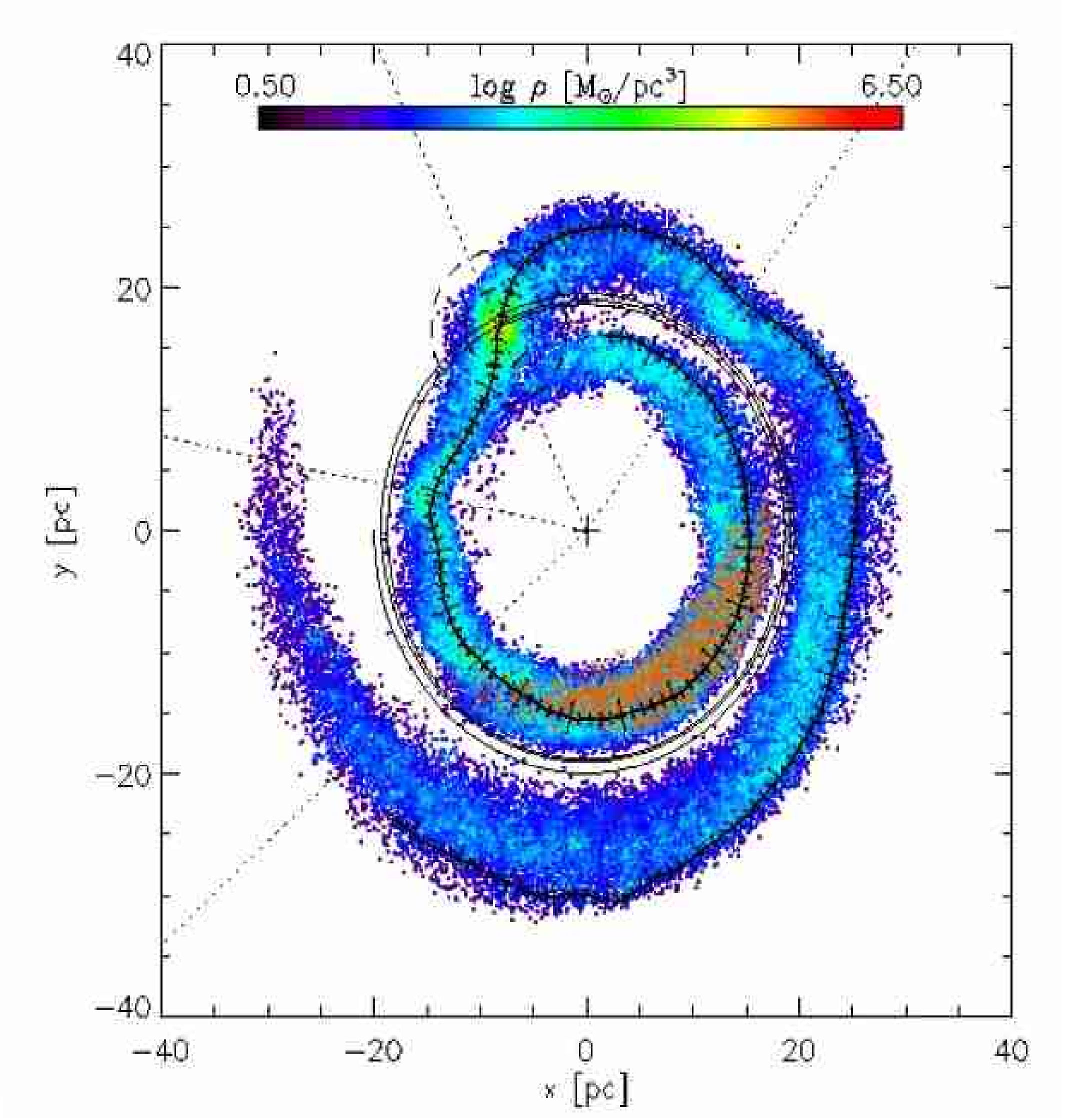
brown tracer particles...

Dissolution (III)

t=1.74 Myr



t=1.30 Myr



Wave phenomenon!

Eigenvector coordinate system

$$\Theta_{jk} = \sum_{i=1}^{N_{nb}} m_i \left(\begin{array}{ccc} \Delta y_i^2 + \Delta z_i^2 & \Delta x_i \Delta y_i & \Delta x_i \Delta z_i \\ \Delta x_i \Delta y_i & \Delta x_i^2 + \Delta z_i^2 & \Delta y_i \Delta z_i \\ \Delta x_i \Delta z_i & \Delta y_i \Delta z_i & \Delta x_i^2 + \Delta y_i^2 \end{array} \right)$$

1) Take neighbor sphere with radius R_{cut}

2) Evaluate averaged physical quantities in the sphere

3) Calculate eigenvalues and eigenvectors of Θ_{jk}

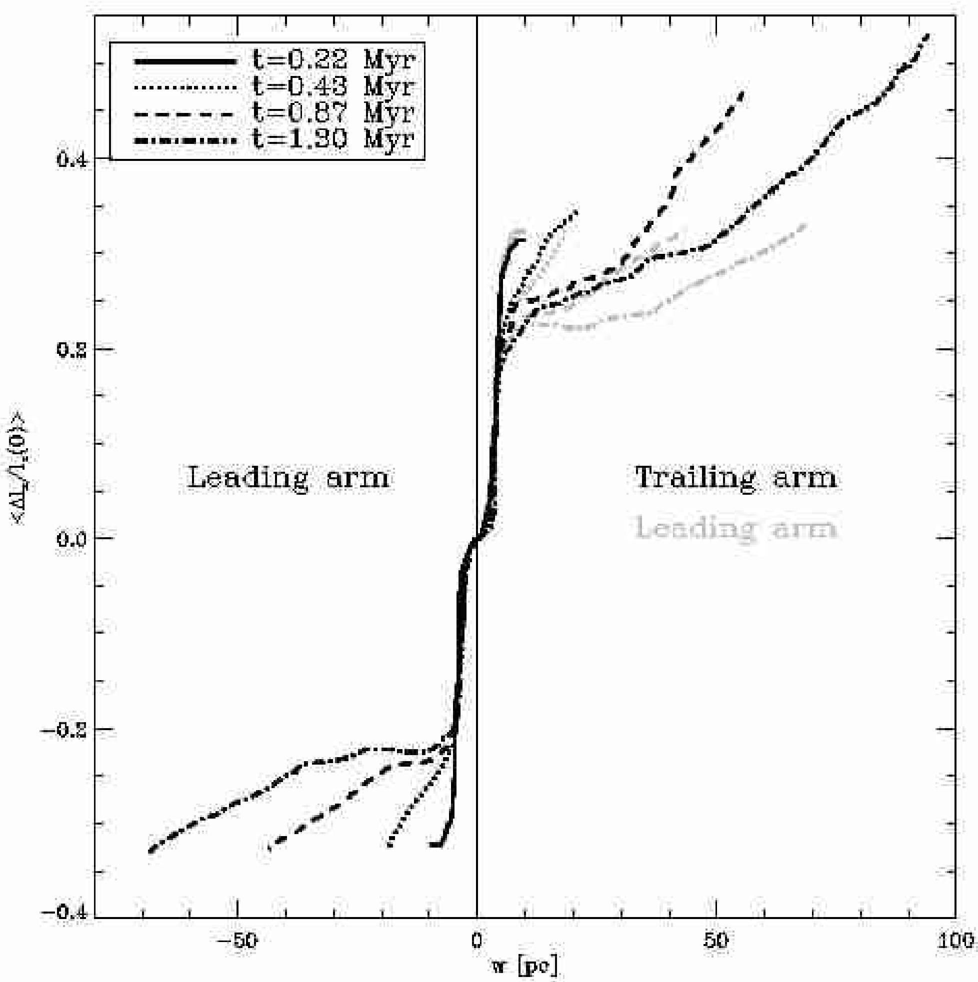
4) Go along maximum eigenvector until stellar density drops below a certain threshold to find new R_{cut}

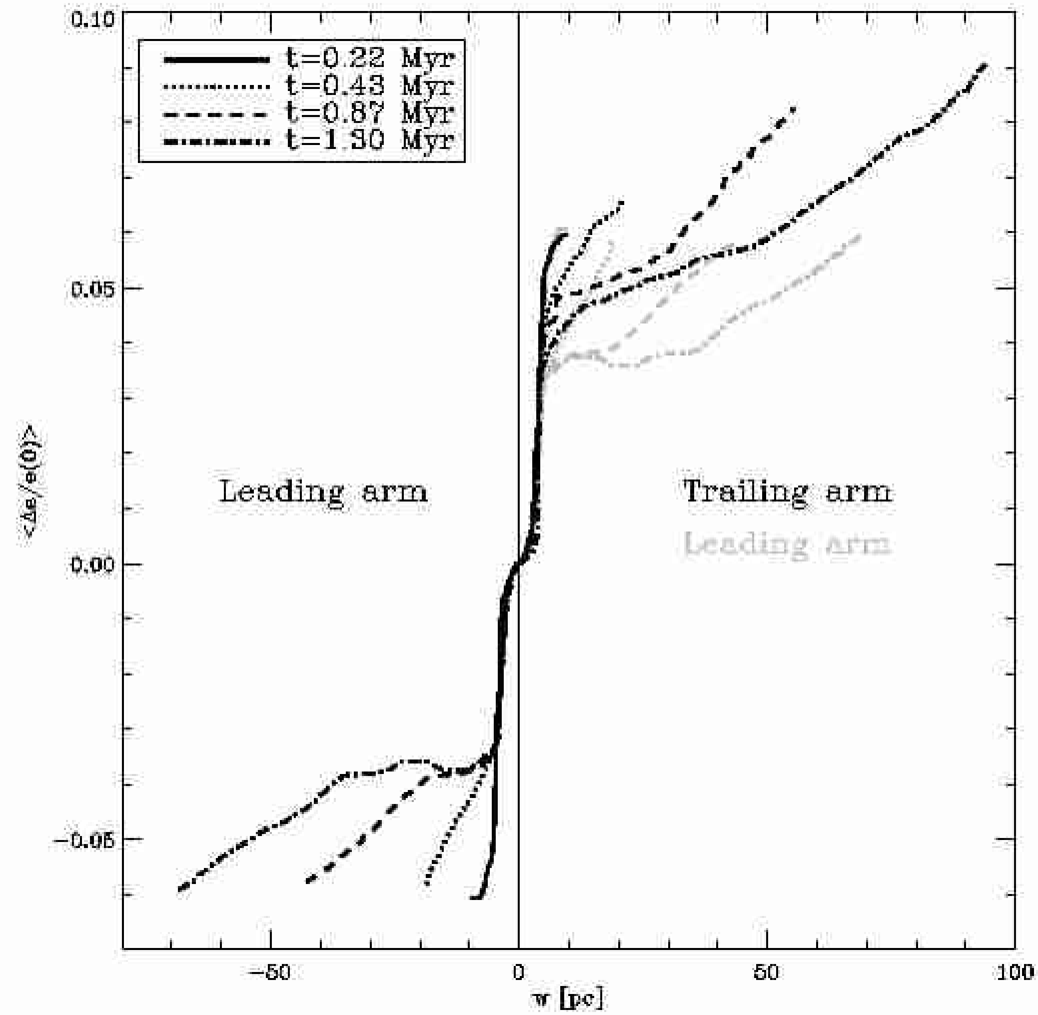
5) Go one step along minimum eigenvector

6) Stop if density has dropped below a certain threshold

7) Start from 1)

Dynamical quantities (I)

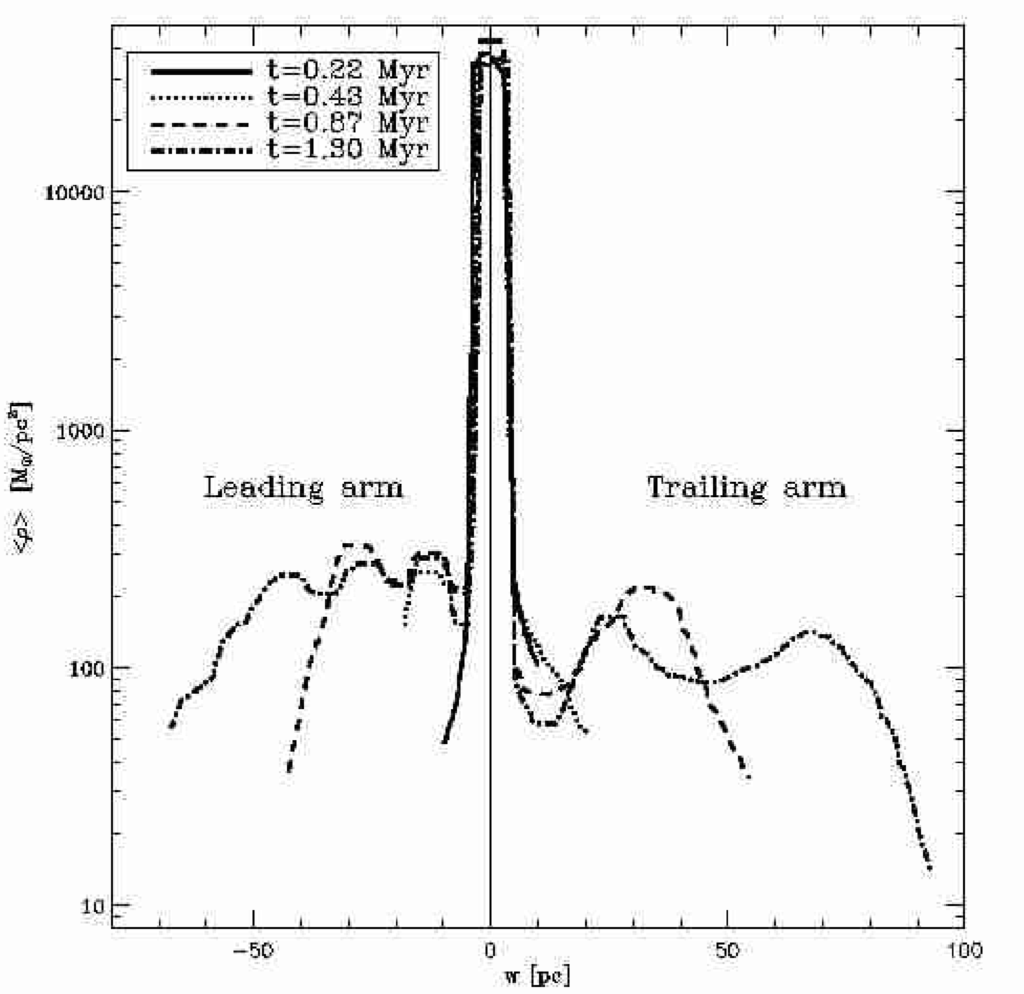


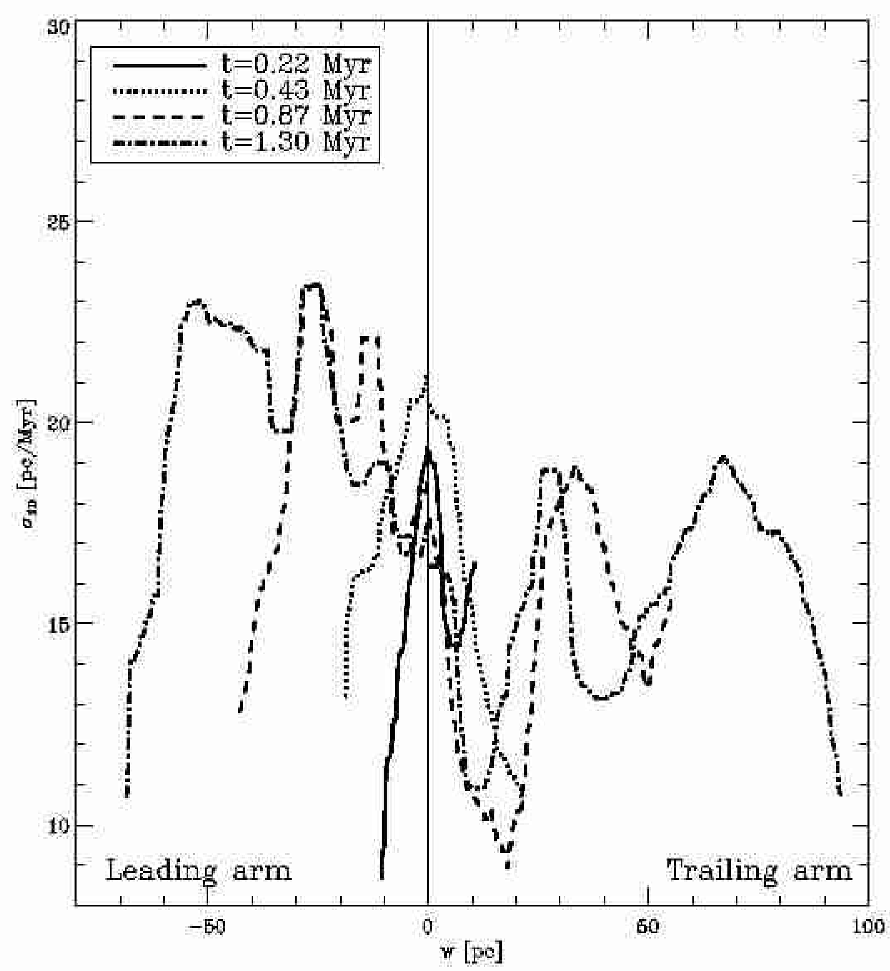


Angular momentum

Energy

Dynamical quantities (II)



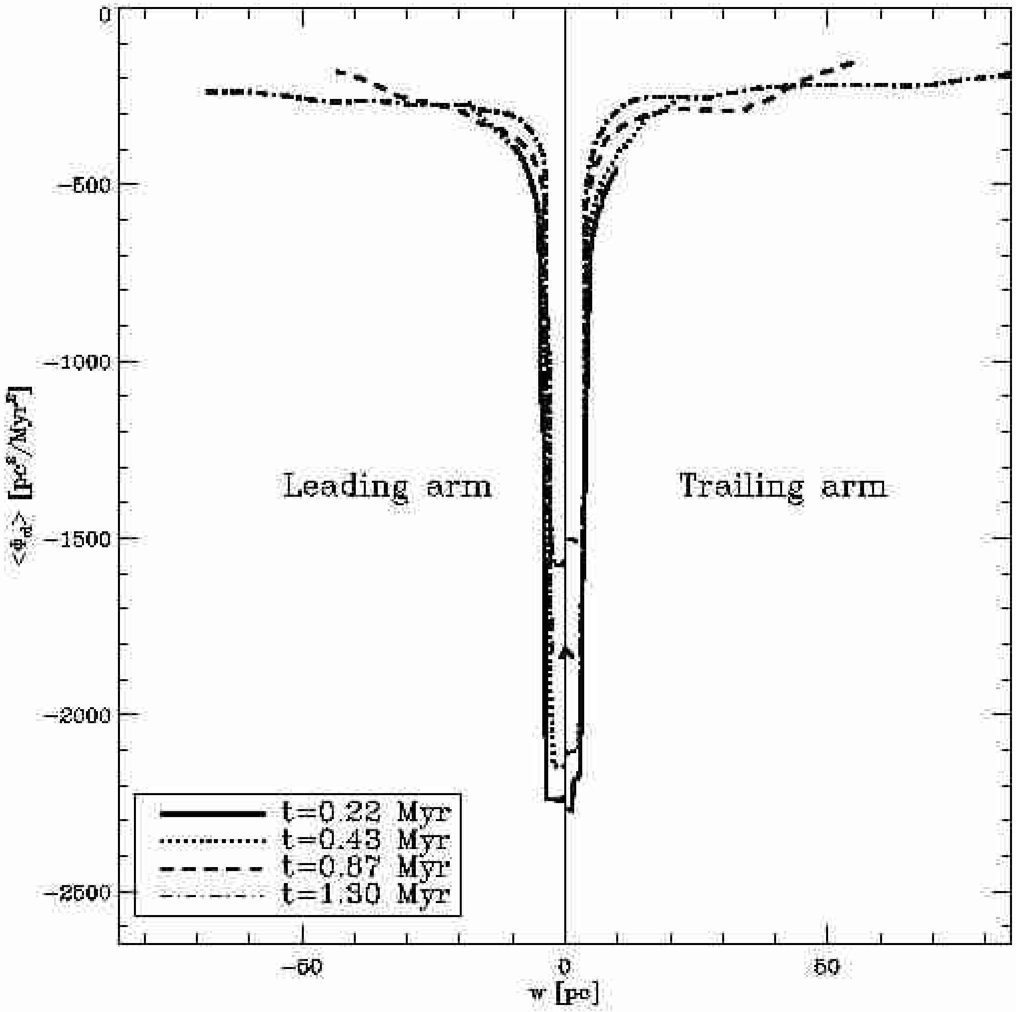


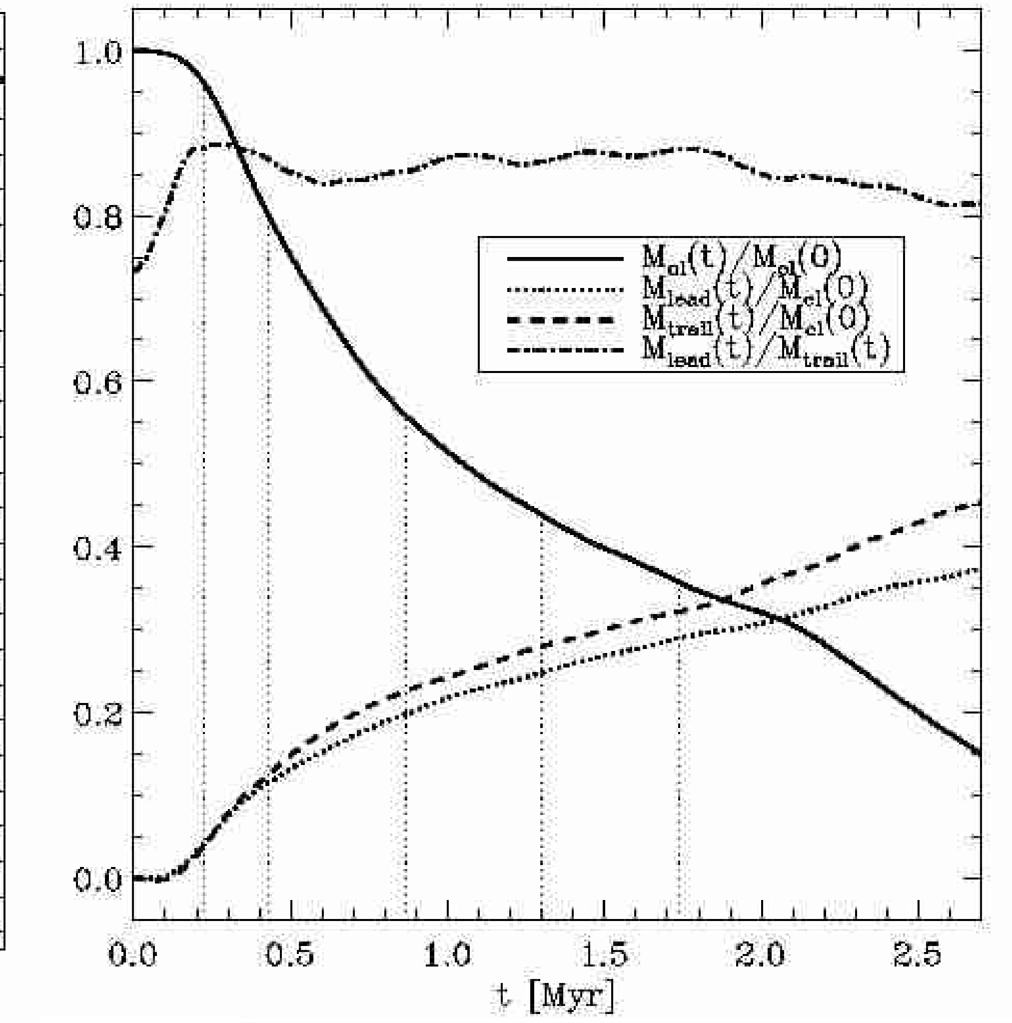
Density

Velocity dispersion

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (\vec{v}_i - \bar{\vec{v}}_i)^2$$

Dynamical quantities (III)

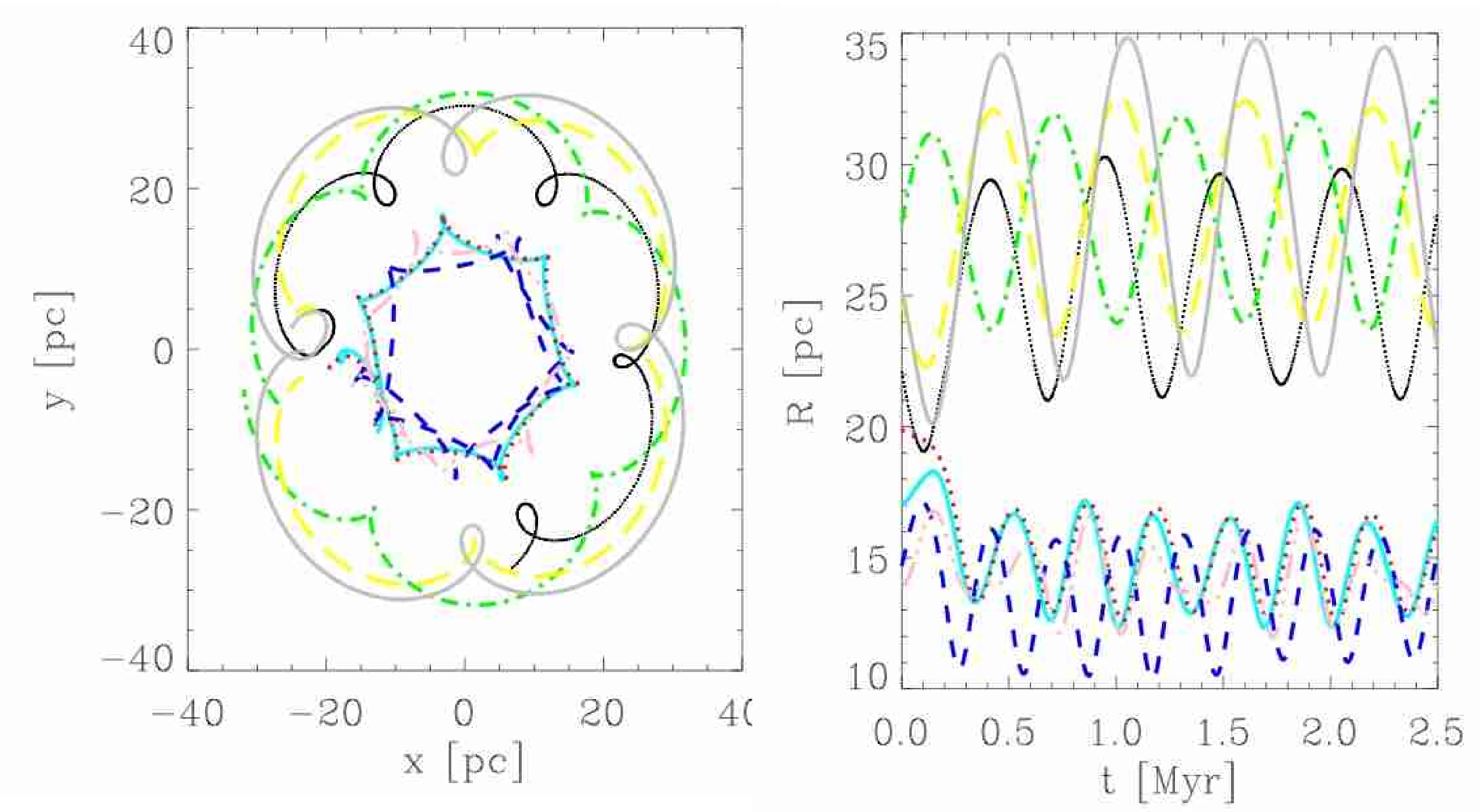




Grav. Potential

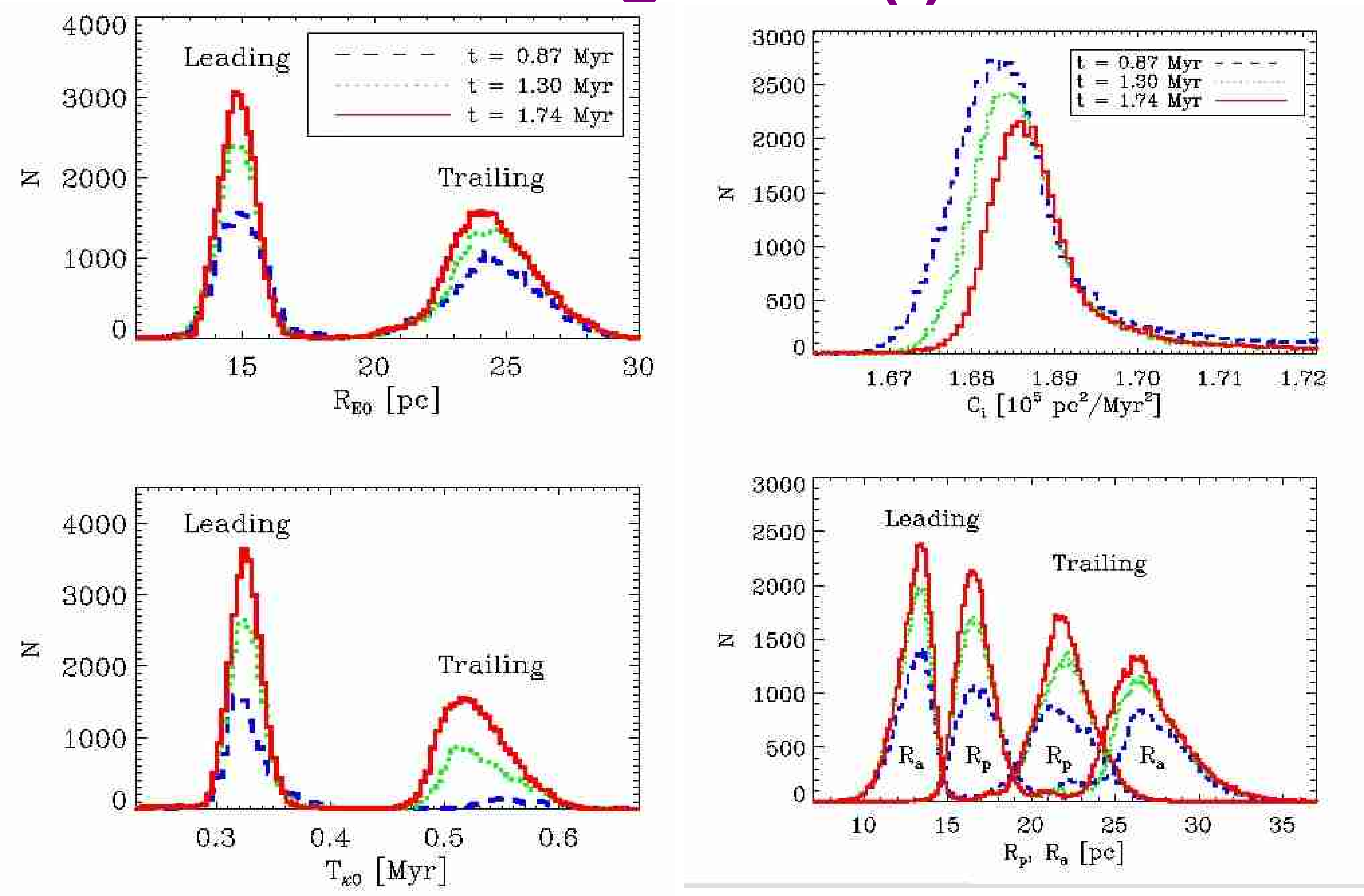
Mass loss...

Cycloids, Epicycles



harmonic oscillation!

Histograms (I)



Analytical theory

$$\Phi(R) \propto R^{\alpha-1}, \quad M(R) \propto R^{\alpha}, \quad \rho(R) \propto R^{\alpha-3},$$

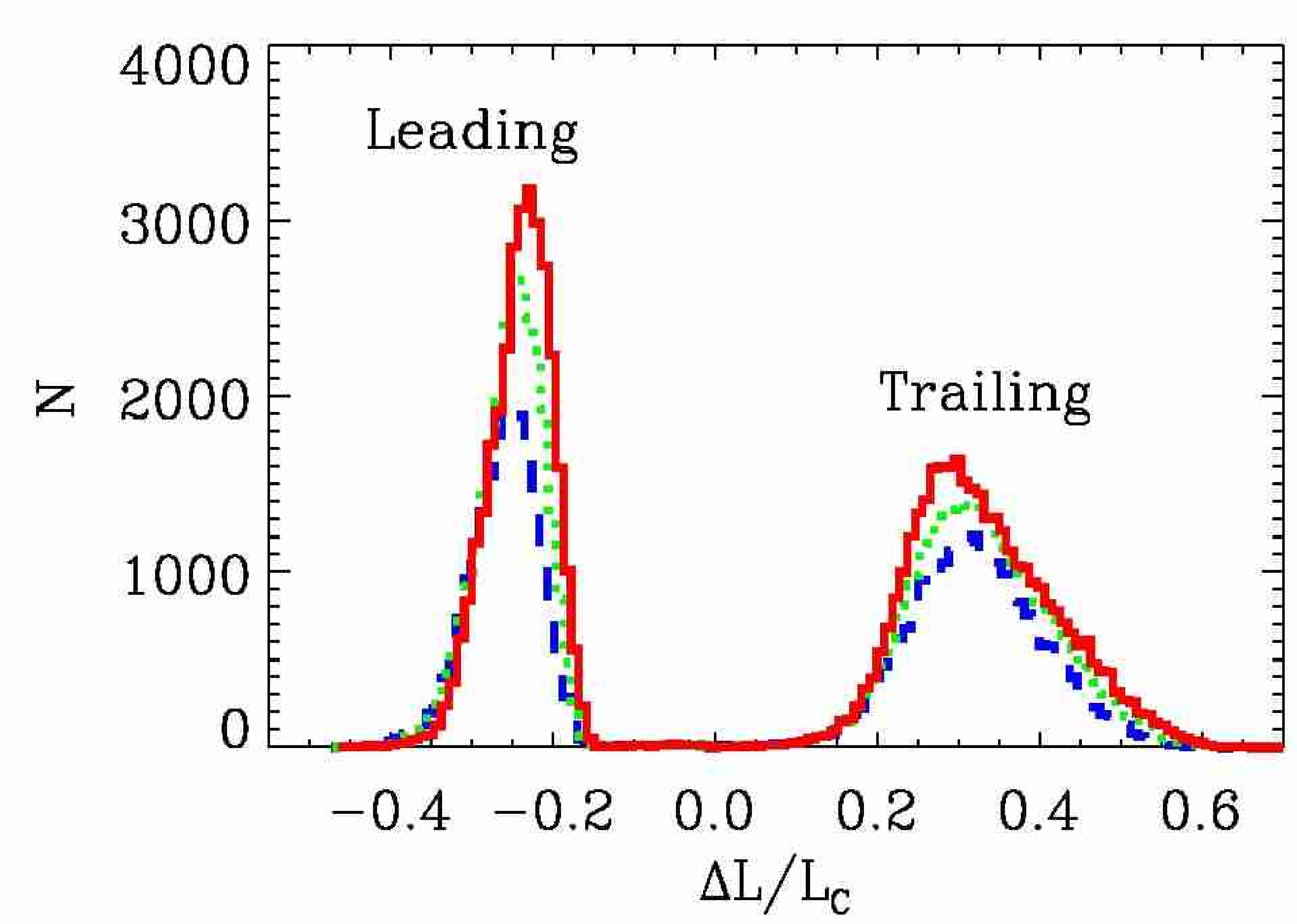
$$\Rightarrow \quad \omega(R) \propto R^{(\alpha-3)/2}, \quad L(R) = \omega R^2 \propto R^{(\alpha+1)/2}$$

$$\beta = \frac{\kappa}{\omega} = \sqrt{2\left[\frac{d\ln\omega}{d\ln R} + 2\right]} = \sqrt{\alpha+1}, \qquad T_\kappa = \frac{2\pi}{\kappa}$$

$$\varphi = T_\kappa \Delta\omega = \frac{2\pi}{\beta}\left(1 - \frac{\omega_C}{\omega}\right) = \frac{2\pi}{\sqrt{\alpha+1}}\left[1 - \left(1 + \frac{\Delta L}{L_C}\right)^{\frac{3-\alpha}{\alpha+1}}\right]$$

$$\Delta\omega = \omega - \omega_C, \quad \Delta L = L - L_C$$

Histograms (II)

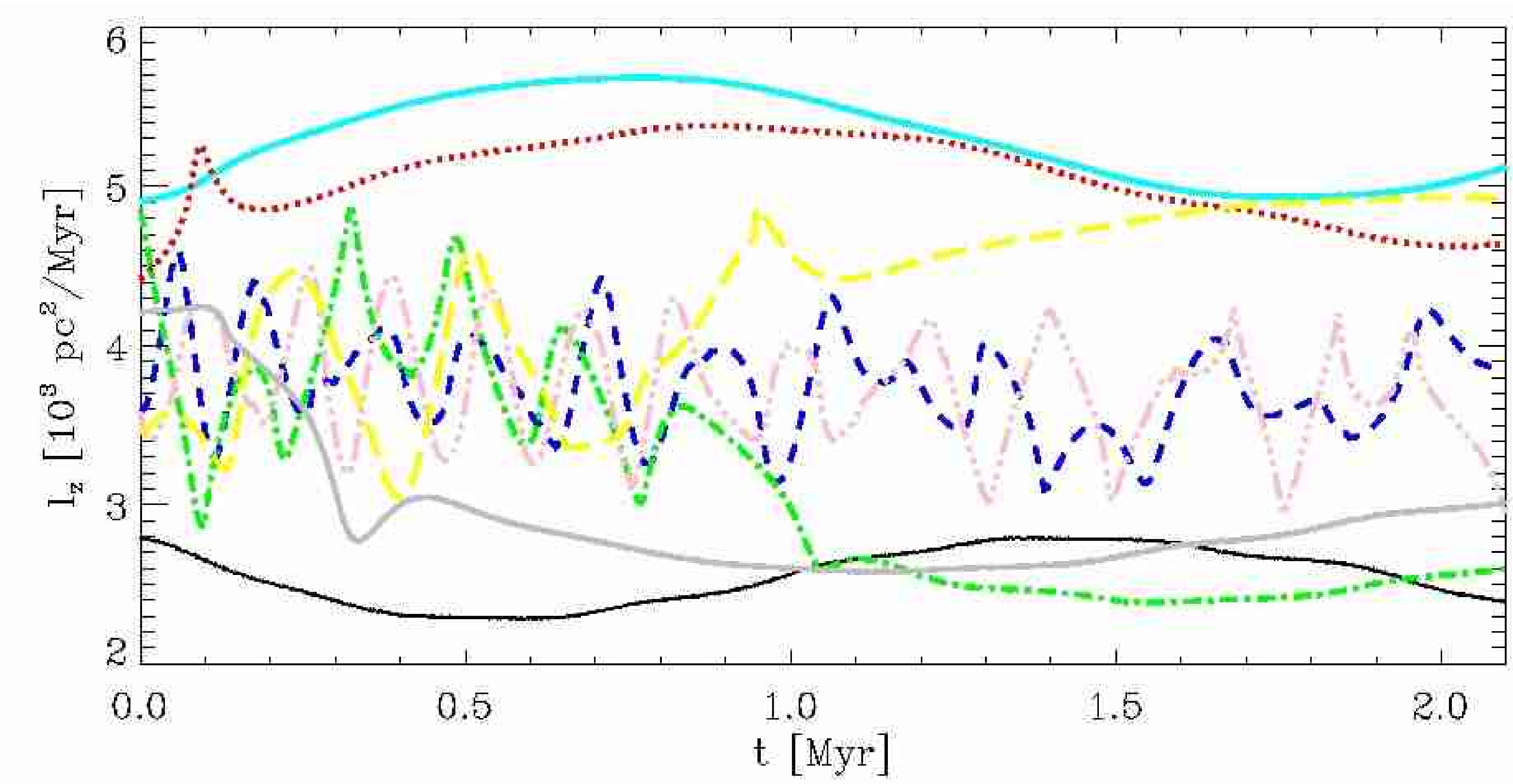


#	t [Myr]	Arm	Clump	$\Delta L/L_C$	$\lvert\varphi_0\rvert$ [deg]	$\lvert\varphi_L\rvert$ [deg]	$\Delta\varphi/\varphi_0$ [%]	$R_C(t)$ [pc]
1	0.87	lead.	1	-0.232	57.5	47.1	18.1	19.0
2	"	lead.	2	-0.273	67.6	55.7	17.6	"
3	"	trail.	1	0.320	68.6	61.9	9.8	"
4	1.30	lead.	1	-0.215	57.5	43.6	24.2	18.8
5	"	lead.	2	-0.227	53.3	46.1	13.5	"
6	"	trail.	1	0.276	68.4	53.6	21.6	"
7	1.74	lead.	1	-0.209	52.6	42.4	19.4	18.6
8	"	lead.	2	-0.205	51.5	41.5	19.4	"
9	"	trail.	1	0.273	64.2	53.0	17.4	"

#	R_{E0} [pc]	R_{EL} [pc]	T_{x0} [Myr]	$T_x(R_{EL})$ [Myr]	$v_t(t-T_{x0})$ [pc]	y_0 [pc]	A_{y0}	A_L
1	14.9	14.9	0.32	0.35	2.67	19.1	1.88	1.54
2	"	14.2	"	0.34	"	22.4	2.20	1.80
3	24.2	24.5	0.55	0.55	2.87	22.7	2.07	1.92
4	14.7	15.1	0.33	0.36	2.40	18.9	2.07	1.54
5	"	14.9	"	0.35	"	17.5	1.91	1.63
6	24.7	23.5	0.51	0.53	2.63	22.4	2.23	1.79
7	14.7	15.0	0.32	0.36	2.21	17.1	2.03	1.63
8	"	15.1	"	0.36	"	16.7	1.98	1.58
9	24.0	23.2	0.51	0.53	2.29	20.8	2.38	2.01

#	$A_{R_{E0}}$	$\Delta A/A_{y0}$ [%]	R_{P0} [pc]	R_{A0} [pc]	$\Delta C/\omega_C^2$ [pc^2]	Δr_L [pc]	R_{PL} [pc]	R_{AL} [pc]
1	1.54	18.1	16.6	13.6	12.83	4.3	19.2	10.6
2	"	18.1	16.6	13.6	"	4.7	18.9	9.5
3	1.81	7.2	21.3	26.2	"	5.0	19.5	29.5
4	1.71	25.6	16.5	13.3	11.27	4.0	19.1	11.1
5	"	14.7	16.5	13.3	"	4.1	19.0	10.8
6	2.24	19.7	22.1	26.2	"	4.5	19.0	28.0
7	1.76	19.7	16.5	13.3	9.71	3.8	18.8	11.2
8	"	20.2	16.5	13.3	"	3.8	18.9	11.3
9	2.36	15.5	21.4	26.0	"	4.3	18.9	27.5

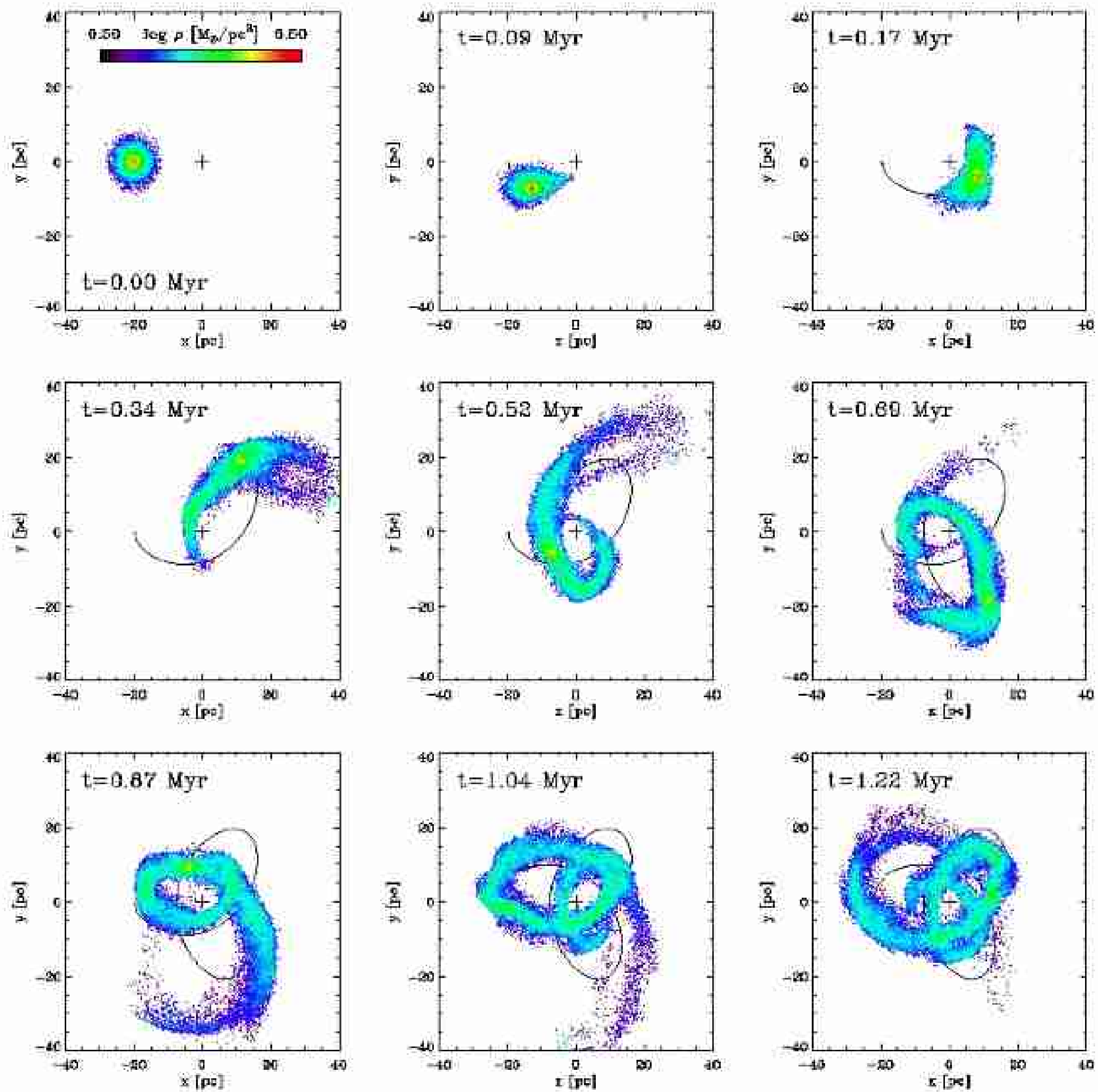
Non-conservation of angular momentum



Movie!

Conclusions

- I have discussed the algorithm of a new parallel N-body program
- Effective potential governs the dynamics in the tidal field
- Lagrange points L_1 and L_2 lie at different energies in the Galactic centre
- Eigensolver can be used to study the structure of tidal arms
- Wave phenomenon in tidal arms can be treated with epicyclic theory


0.50 log ρ [M☉/pc³] 6.50
t=0.00 Myr
t=0.09 Myr
t=0.17 Myr
t=0.34 Myr
t=0.52 Myr
t=0.69 Myr
t=0.87 Myr
t=1.04 Myr
t=1.22 Myr
x [pc]
y [pc]